

Railway Ticket Booking Management System

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STRUCTURAL SYMBOLS



Entity

Real-world object / record set (e.g., Passenger, Train)



Weak Entity

Existence-dependent on owner entity (e.g., Seat/Berth)



Relationship

Association connecting 2 or more entities (e.g., Books)



Weak Relationship

Double diamond linking weak entity to owner entity



Attribute to Entities

Links attribute to entity sets or entity sets to relationship sets

ATTRIBUTE CLASSIFICATIONS



Key Attribute

Uniquely identifies an entity instance (e.g., PNR_No)



Simple / Atomic

Indivisible basic attribute value (e.g., Fare, Gender)



Composite

Subdivided into components (e.g., Name → FName, LName)



Multi-Valued

Can hold multiple distinct values (e.g., {Phone_No})



Derived

Calculated from other stored data (e.g., Age from DoB)

PARTICIPATION & ADVANCED



Total Participation

Mandatory: Every entity instance must participate



Partial Participation

Optional: Entity instances may or may not participate



Cardinality Ratio

Defines association limits (1:1, 1:N, M:N mapping)



Ternary Relation

Simultaneous 3-way link connecting three entity sets

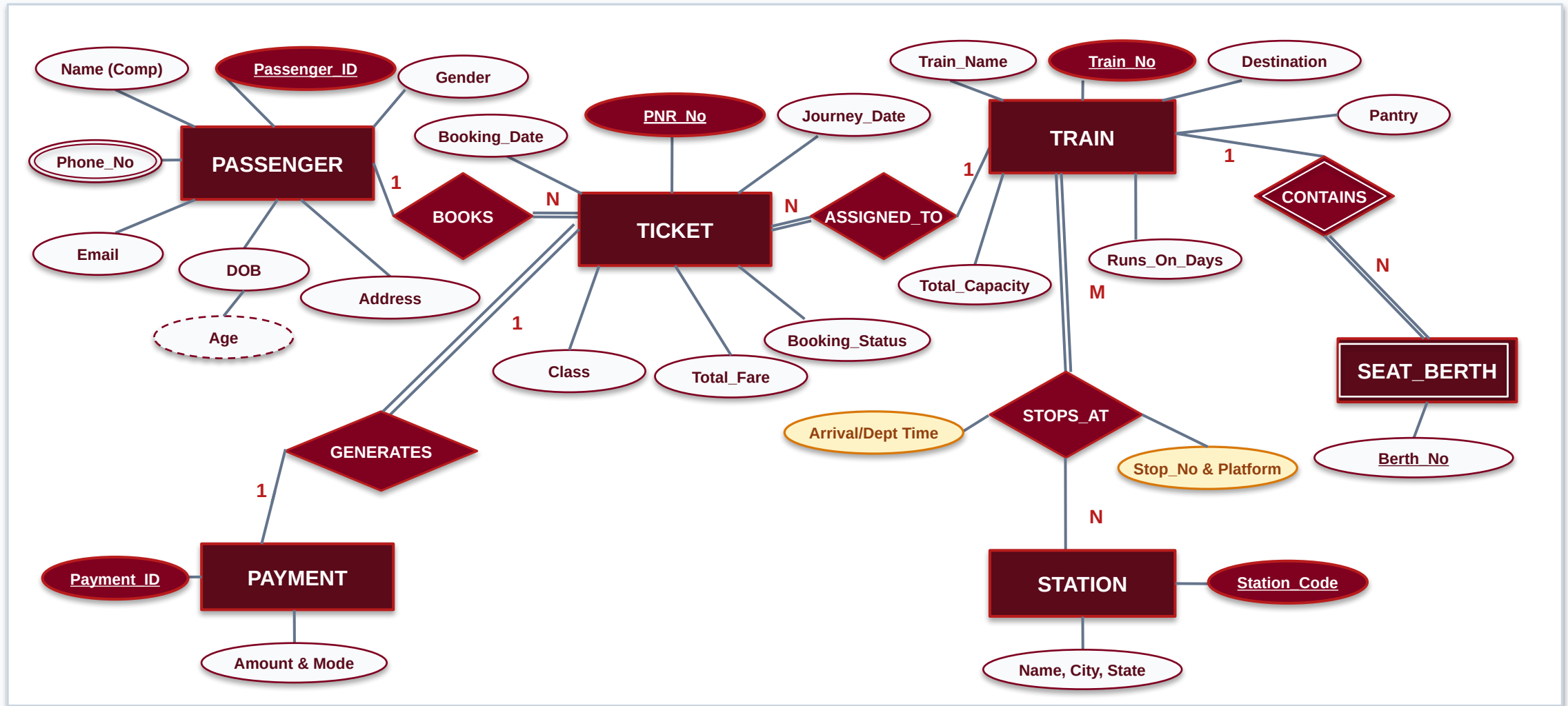


Specialization (ISA)

Superclass/subclass inheritance hierarchy structure

ER Diagram

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Relational Database Schema

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1. Entity Sets & Primary Key Architecture

- **Candidate & Primary Keys:** PNR_No acts as the primary key for booking transactions; Train_No
- **Weak Entity Identification:** Seat_Berth is dependent on Train, identified via composite attributes (Train_No + Berth_No).
- **Derived & Multivalued Attributes:** Passenger Age is dynamically derived from Date of Birth (DoB), multi-valued phone numbers are decomposed into PASSENGER_PHONE.

2. Relationship Cardinality & Route Network Mapping

- **1:N Booking Links:** One registered passenger can book multiple tickets, while each individual ticket is legally owned by a single account.
- **M:N Complex Route Association:** Train to Station (STOPS_AT) is Many-to-Many, trains visit multiple stations and stations service multiple distinct train lines.
- **1:1 Payment Ledger & Ternary Links:** Every confirmed TICKET strictly generates 1 financial payment record; reservation atomically links passenger, train, and berth.

3. Descriptive Attributes, Participation & Integrity Constraints

- **Descriptive Link Attributes:** Arrival_Time, Departure_Time, and Platform_No describe the STOPS_AT association and migrate into the TRAIN_SCHEDULE junction table.
- **Total vs Partial Participation:** Ticket has mandatory total participation, passenger participation is partial.

Thank You!